

# Myeongju Cha

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## Contact

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## Research Interests & Objective

- Powered lower-limb prosthesis control
  - Robotics & embedded system design
  - Gait phase estimation and control algorithms
  - **Goal:** Develop wearable robotic systems enabling natural locomotion in diverse environments
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## Education

- **2024 – Present** Ph.D. in Mechanical Engineering, GIST, Korea
  - **2022 – 2024** M.S. in Mechanical Engineering, GIST, Korea
  - **2016 – 2021** B.S. in Mechanical Engineering, GIST, Korea
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## Research Experience

- **2023 – Present Exoskeleton Control Box Design & Control Implementation**
    - Implemented gravity-compensator + PD trajectory-tracking control for walking, standing, and sitting
    - Developing a wearable rehabilitation system for hemiplegic patients
  - **2023 – Present Free-Energy-Principle-Based Knee Assist Robot**
    - Applied FEP control to a single-motor knee brace; benchmarking against PD and deep-learning controllers
    - Designed backpack-type controller and experimental protocol
  - **2023 – 2024 Portable Knee CPM Device**
    - Designed PD + DOB controller to compensate friction/disturbances for smooth CPM motion
    - Integrated detachable hardware (in collaboration with structure-design team) and firmware
  - **2024 – Present Powered Prosthesis Stair-Descent Control Framework**
    - Led CAN-based embedded control system design and experiments
  - **2019 Sound-Direction-Sensing Robot** — Microphone-array signal processing & localization algorithm
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## Conferences & Workshops

- **2025 International Conference on Control, Robotics & Systems (ICROS)** — *Design and Trajectory Tracking Control of a Customizable Active Prosthesis Leg* — **Poster**

- **2024 International Conference on Control, Robotics & Systems (ICROS)** — *Phase-Variable Generation for Stair-Descent State Prediction* — **Oral**
  - **2023 Joint Conference of the Korean Society of Biomechanics & Korean Society of Sports Biomechanics (KSB-KSSB)** — *Phase-Variable Generation for Stair-Descent Gait State Estimation* — **Oral**
  - **2022 KSME IT Convergence Conference** — *Control Framework for Powered Prosthesis on Inclined Walking* — **Oral**
  - **2021 ICROS** — *Dynamic Modeling & Control of a Compass-Gait Walker* — **Poster**
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## Publications

1. **M Cha**, P Hur, "Estimation of Gait Phase of Human Stair Descent Walking Based on Phase Variable Approach," *IEEE Robotics and Automation Letters*, **2025**, DOI: 10.1109/LRA.2025.3579634.
  2. **M Cha**, P Hur, "Phase-Variable Generation for Stair-Descent State Prediction," *Proc. ICROS*, **2024**, pp. 406-407.
  3. **M Cha**, W Hong, P Hur, "Control Framework for Powered Prosthesis on Inclined Walking," *Proc. KSME*, **2022**, p. 80.
  4. **M Cha**, S Moon, G Cho, P Hur, "Dynamic Modeling & Control of a Compass-Gait Walker," *Proc. ICROS*, **2021**, pp. 577-578.
  5. S Moon, **M Cha**, G Cho, P Hur, "Optimal Control of Compass-Gait Walker via Direct Collocation," *Proc. ICROS*, **2021**, pp. 579-580.
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## Teaching & Assistant Experience

- **2024 F & 2025 S Motor-Control Boot-Camp** (2 iterations) — Planner & lead instructor
  - **2022 Spring Signals & Systems in Mechanical Eng.** — Teaching Assistant, **monthly** recitations & grading
  - **2021 Fall Dynamics** — Teaching Assistant, **monthly** recitations & grading
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## Awards

- **Best Paper Award**, ICROS 2024 — *Phase-Variable Generation for Stair-Descent State Prediction*
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## Volunteer Experience

- **2018** High-School Tutoring Program (Korea Student Aid Foundation) — One-month math & science teaching, 8-member team
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## Skills & Competencies

### Hardware / Embedded

STM32 (M-Series, H755) · Raspberry Pi 4 · Jetson Nano · Teensy 4.1  
CAN · I2C · SPI bus design & integration

### Software / Programming

C/C++ · Python · ROS 1 & 2 · MATLAB/Simulink

Real-time firmware · multi-threaded Linux applications

### **Data Analysis & Statistics**

IMU / FSR data preprocessing & fusion

RMSE ·  $R^2$  · one-way & repeated-measures ANOVA